Date 27/09/22

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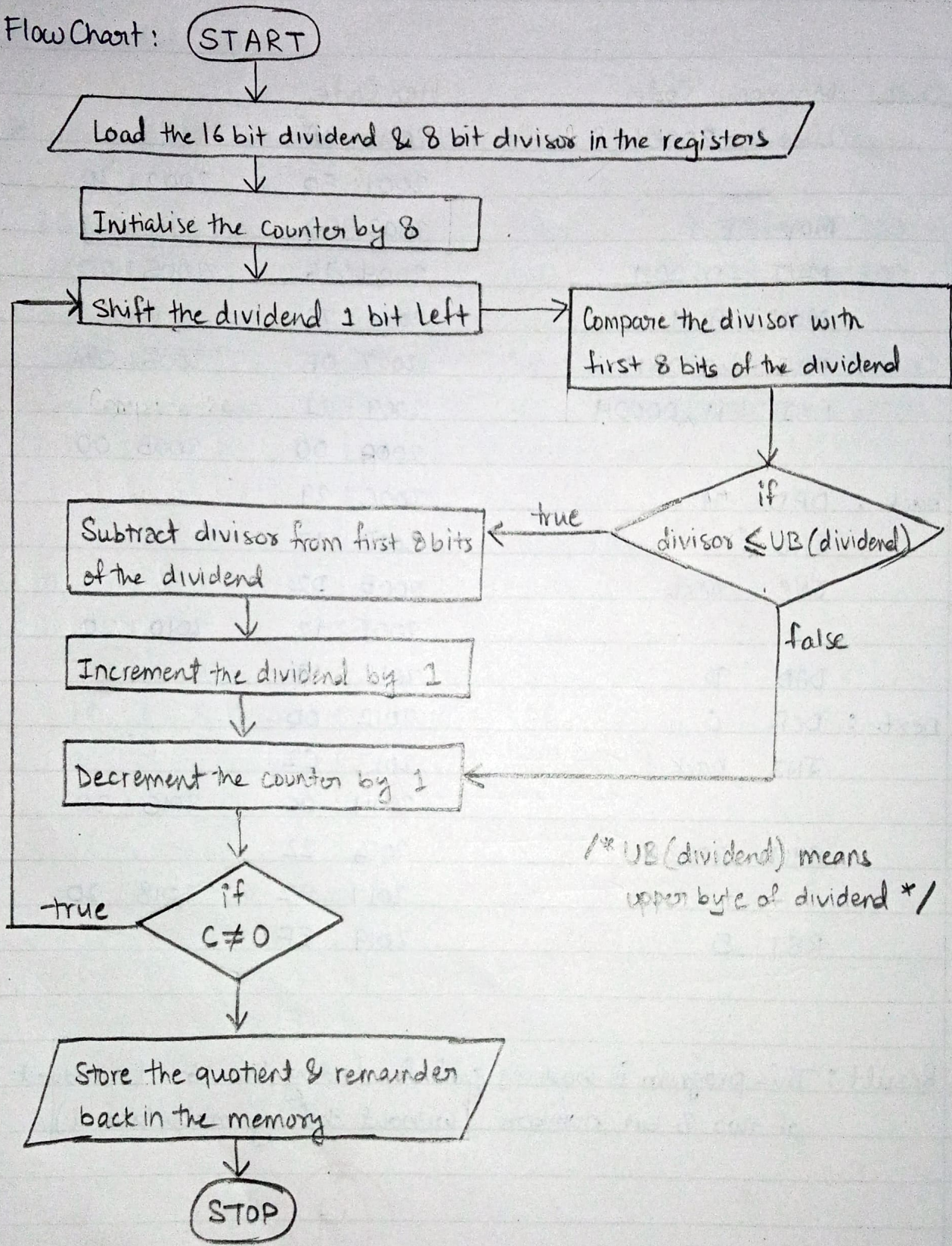
- Aim: Write a program in assembly language of 8085 to find the Quotient & Remainder after dividing a 16 bit number by an 8 bit number. (by shifting dividend)

- Registers Used: HL pair : to store 16 bit Dividend (in beginng)
  B reg : to store 8 bit divisor
  Accumulator : to compare values.
  C reg : to use as a counter in loop
  H reg : to store the remainder (in end)
  L reg : to store the quotient (in end)

- Algorithm: (i) start and initiate the address of the program.
(ii) Load the HL pair with the 16 bit Dividend
(iii) Load the B reg with 8 bit divisor, and initiate Creg with value 08H.
(iv) Left shift the dividend stored in HL by 1 bit.
(v) compare the divisor with first 8 bits of the dividend i.e. H reg.
(vi) if divisor is greater than the H value, jump to step (viii)
(vii) else sub B from H (H ← H-B) & increment L value by 1.
(viii) decrement counter in C by 1, if C is not zero jump to step (iv)
(ix) now H register contains the remainder & L register contains the quotient.
(x) store the quotient & the remainder in the memory
(xi) stop.

Teacher's Signature : _____

• Flow Chart:


START
Load the 16 bit dividend & 8 bit divisor in the registers
Initialise the counter by 8
Shift the dividend 1 bit left
Compare the divisor with first 8 bits of the dividend
if divisor ≤ UB(dividend)
true
false
Subtract divisor from first 8 bits of the dividend
Increment the dividend by 1
Decrement the counter by 1
if C≠0
true
Store the quotient & remainder back in the memory
STOP
/* UB(dividend) means upper byte of dividend */

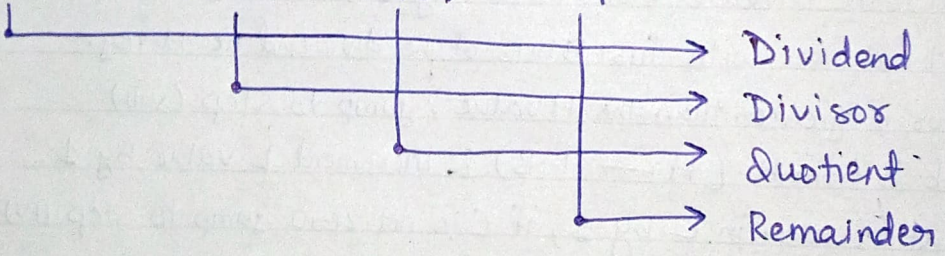
- Observation Table:

Memory Addr	Before	After	Decimal Equiv.
2050 H	DB	DB	69DB → $27099_{(10)}$
2051 H	69	69	
2052 H	B9	B9	B9 → $185_{(10)}$
2053 H	00	92	92 → $146_{(10)}$
2054 H	00	59	59 → $89_{(10)}$

- Calculations:

Dividend = 69DB H = 27099
Divisor = B9 H = 185

27099 = (185 × 146) + 89



Quotient = 92 H = $146_{(10)}$
Remainder = 59 H = $89_{(10)}$

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- Code

	Mnemonic Code		Hex code	
	LHLD	2050H	2000: 2A	
			2001: 50	2002: 20
	LDA	2052H	2003: 3A	
			2004: 52	2005: 20
	MOV	B,A	2006: 47	
	MVI	C,08H	2007: 0E	2008: 08
back:	DAD	H	2009: 29	
	MOV	A,H	200A: 7C	
	SUB	B	200B: 90	
	JC	next	200C: DA	
			200D: 11	200E: 20
	MOV	H,A	200F: 67	
	INR	L	2010: 2C	
next:	DCR	C	2011: 0D	
	JNZ	back	2012: C2	
	[illegible]		2013: 09	2014: 20
	SHLD	2053H	2015: 22	
	[illegible]		2016: 53	2017: 20
	RST	5	2018: EF	

- Result : This program is working good for finding the quotient & remainder upon dividing a 16 bit number by an 8 bit number. (by shifting dividend).

Teacher's Signature : _____